

# HAO ZOU

(631) 809-8931 [◇ haozou-official.github.io](https://haozou-official.github.io) [◇ hz2999@columbia.edu](mailto:hz2999@columbia.edu)

## EDUCATION

---

### Columbia University

*MS in Computer Science, Machine Learning Track*

New York, NY

09/2024 - 05/2026

### University of Minnesota, Twin Cities

*BS in Computer Science*

Minneapolis, MN

09/2019 - 05/2023

Research Interests: Diffusion Models, Inference-Time Algorithms for LLMs, LLM Reasoning, LLM Agents

## PUBLICATIONS

---

### No Compute Left Behind: Rethinking Reasoning and Sampling with Masked Diffusion Models

Zachary Horvitz, Raghav Singhal, **Hao Zou**, Carles Domingo-Enrich, Zhou Yu, Rajesh Ranganath, Kathleen McKeown

Under review, ICLR 2026. [\[arxiv\]](#)

### A Survey of Diffusion Models in Natural Language Processing

**Hao Zou**, Zae Myung Kim, Dongyeop Kang

Submitted, EMNLP 2023. [\[arxiv\]](#)

### Debiasing Language Models for In-Context Learning Using a Causal Inference-Inspired Method

**Hao Zou**, Karin de Langis, Dongyeop Kang, Yohan Jo

Submitted, EACL 2023. [\[paper\]](#)

### You Make me Feel like a Natural Question: Training QA Systems on Transformed Trivia Questions

Tasnim Kabir, Yoo Yeon Sung, Saptarashmi Bandyopadhyay, **Hao Zou**, Abhranil Chandra, Jordan Boyd-Graber  
EMNLP 2024. [\[paper\]](#)

### Improving Question Answering with Generation of NQ-like Questions

Saptarashmi Bandyopadhyay, Shraman Pal, **Hao Zou**, Abhranil Chandra, Jordan Boyd-Graber

Submitted, MRQA 2021. [\[paper\]](#)

## RESEARCH EXPERIENCE

---

### NLP Lab, Columbia University

*Graduate Research Assistant*

02/2025 - Present

*Advisors: Kathleen McKeown & Zhou Yu*

#### • Rethinking Reasoning in Masked Diffusion Language Models (MDLMs)

- Investigated the behavior of any-order vs autoregressive decoding in LLaDA across GSM8K, MATH500, HumanEval, Sudoku, and MT. Showed that simple left-to-right decoding can match or exceed any-order methods on structured reasoning tasks, motivating a redesign of inference-time compute allocation.
- Contributed experiments demonstrating that MDLMs can infill structured reasoning templates, enabling answer-conditioned posterior sampling of reasoning traces, uncertainty-based early exit, and inference-time reasoning validation without external verifiers.

#### • Revision-Based Diffusion Decoding for Summarization

- Developing discriminator-guided revision strategies for MDLMs using ReMDM + Informed Corrector (IC). Designed refill policies that combine entropy, discriminator signals, and FK-steering rewards to revise low-faithfulness spans during decoding.

#### • OSWorld-V2 Benchmark for Long-Horizon LLM Agents

- Designed and implemented complex, high-fidelity desktop tasks (50–200 steps) for evaluating autonomous LLM agents in multi-application settings (browser, spreadsheets, docs, email, system utilities).
- Contributing to a library of agent-challenging tasks with multi-step dependencies, external website integration, and long-horizon planning requirements for OSWorld-V2 benchmark release.

- Developed a benchmarking pipeline comparing diffusion and flow-matching training paradigms under different prediction objectives across diverse architectures.
- Introduced a triangle-distribution-based training objective improving stability and generation quality over Gaussian baselines.

- Developed a method driven by causal inference to measure true causal effect of input text on potential labels, boosted Pre-trained Language Models (PLMs) accuracy across various tasks by up to 22% points.
- First-authored an in-depth survey analyzing diffusion model formulations for NLP, including forward–reverse processes, denoising objectives, parallel generation properties, token-level control mechanisms, and robustness characteristics. Benchmarked diffusion approaches against autoregressive Transformer models and outlined research directions for scaling multimodal and large-parameter diffusion LMs.

- Developed a pipeline for transforming complex trivia questions into information-seeking forms using syntactic simplification and question rewriting models, enabling more robust passage retrieval and downstream QA performance.
- Fine-tuned a Dense Passage Retriever (DPR) on the rewritten questions to better align retriever embeddings with the transformed query distribution, yielding an 18% accuracy gain in cross-domain QA evaluation.
- Conducted fine-grained linguistic and stylistic analyses on 2k+ generated questions to characterize generalization patterns across domain shifts, syntactic variations, and semantic abstraction levels.

WORK EXPERIENCE

- Optimized inference cost for IBM platforms by proposing a Johnson-Lindenstrauss-based KV-cache quantization approach integrated with sparsity pruning, achieving 50% memory reduction and 2× inference speedup.
- Benchmarked quantized attention mechanisms across multiple LLM families.

- Developed a multi-style controllable generation pipeline using PPO and novel fusion methods, improving human-judged style alignment by 31.2%.

ACADEMIC SERVICE

Conference Reviewer ACL, EMNLP, EACL

SKILLS

|                  |                                                                                |
|------------------|--------------------------------------------------------------------------------|
| Machine Learning | Diffusion Models, MDLMs, Flow Matching, RLHF, PPO, SMC Sampling                |
|                  | LLM Inference-Time Algorithms                                                  |
| NLP              | Summarization, Reasoning Models, Faithfulness Metrics, RAG, Question Answering |
| Programming      | Python, PyTorch, CUDA, C/C++, Linux, Git, Docker                               |
| Specialties      | LLM Agents, Long-Horizon Planning, Evaluation Benchmarking, Causal Inference   |